

Commerce XI Maths App

Miscellaneous Exercise 1

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 33

Question 1 - Q1(i)

The data gives number of accidents per day on a railway track,
4, 2, 3, 5, 6, 3, 4, 1, 2, 3, 2, 3, 4, 3, 2.

Compute Q_2

Correct answer: 3

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 3.

Question 2 - Q1(ii)

The data gives number of accidents per day on a railway track,
4, 2, 3, 5, 6, 3, 4, 1, 2, 3, 2, 3, 4, 3, 2.

Compute P_{17}

Correct answer: 2

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 2.

Question 3 - Q1(iii)

The data gives number of accidents per day on a railway track,
4, 2, 3, 5, 6, 3, 4, 1, 2, 3, 2, 3, 4, 3, 2.

Compute D_7

Correct answer: 4

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 4.

Question 4 - Q2(i)

The distribution of daily sales of shoes (size wise) for 100 days from a certain shop is :

Shoe size	2	3	4	5	6	7	8
No. of days	14	20	13	19	13	13	8

Compute Q_2 .

Correct answer: 5

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 5.

Question 5 - Q2(ii)

The distribution of daily sales of shoes (size wise) for 100 days from a certain shop is :

Shoe size	2	3	4	5	6	7	8
No. of days	14	20	13	19	13	13	8

Compute D_1 .

Correct answer: 2

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 2.

Question 6 - Q2(iii)

The distribution of daily sales of shoes (size wise) for 100 days from a certain shop is :

Shoe size	2	3	4	5	6	7	8
No. of days	14	20	13	19	13	13	8

Compute P_{95} .

Correct answer: 8

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 8.

Question 7 - Q3

Ten students appeared for a test in Mathematics and Statistics and they obtained the marks as follows:

Subject	Marks
Mathematics	42, 38, 36, 32, 23, 25, 35, 37, 25, 23
Statistics	22, 26, 29, 34, 50, 45, 23, 28, 32, 36

If the median will be the criteria, in which subject, the level of knowledge of the students is higher?

Options

- A. Knowledge of Mathematics is higher than Statistics.
- B. Knowledge of Statistics is higher than Mathematics.
- C. Both subjects have the same level of knowledge.

Correct answer: Knowledge of Mathematics is higher than Statistics.

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is Knowledge of Mathematics is higher than Statistics..

Question 8 - Q4

In the frequency distribution of families given below, the number of families corresponding to expenditure group 2000–4000 is missing from the table. However value of 25th percentile is 2880. Find the missing frequency x.

Expenditure (in ₹)	0–2000	2000–4000	4000–6000	6000–8000	8000–10000
No. of families	14	x	39	7	15

Correct answer: 25

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 25.

Question 9 - Q5(i)

Calculate Q_1 for the following data :

Mid values	25	75	125	175	225	275
Frequency	10	70	80	100	150	90

Correct answer: 128.125

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 128.125.

Question 10 - Q5(ii)

Calculate D_6 for the following data :

Mid values	25	75	125	175	225	275
Frequency	10	70	80	100	150	90

Correct answer: 213.33

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 213.33.

Question 11 - Q5(iii)

Calculate P_{15} for the following data :

Mid values	25	75	125	175	225	275
Frequency	10	70	80	100	150	90

Correct answer: 96.43

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 96.43.

Question 12 - Q6

Daily income for a group of 100 workers are given below,
 P_{30} for this group is ₹110. Calculate missing frequencies x, y.

Daily income (in ₹)	0-50	50-100	100-150	150-200	200-250
No. of workers	7	x	25	30	y

Options

- A. 18, 20
- B. 20, 18
- C. 17, 21
- D. 22, 16

Correct answer: 18, 20

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 18, 20.

Question 13 - Q7

The distribution of a sample of students appearing for a C.A. examination is given below,
Help C.A. institute to decide cut off marks for qualifying the examination,
when 3% students pass the examination.

Marks	0-100	100-200	200-300	300-400	400-500	500-600
No. of students	130	150	190	220	280	130

Correct answer: 575

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 575.

Question 14 - Q8(i)

Determine graphically the value of median for the data given below :

Class	10-15	15-20	20-25	25-30	30-35	35-40	40-45
Frequency	8	14	8	25	15	14	6

Correct answer: 29

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 29.

Question 15 - Q8(ii)

Determine graphically the value of D_3 for the data given below :

Class	10-15	15-20	20-25	25-30	30-35	35-40	40-45
Frequency	8	14	8	25	15	14	6

Correct answer: 25

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 25.

Question 16 - Q8(iii)

Determine graphically the value of P_{35} for the data given below :

Class	10-15	15-20	20-25	25-30	30-35	35-40	40-45
Frequency	8	14	8	25	15	14	6

Correct answer: 26

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 26.

Question 17 - Q9

The I.Q. test of 500 students of a college is given below,

Find graphically the number of students whose I.Q. is more than 55 graphically.

I.Q.	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No. of students	41	52	64	180	67	45	40	11

Correct answer: 256

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 256.

Question 18 - Q10

Draw an ogive for the following distribution.

Determine the median graphically and verify your result by mathematical formula.

Height (in cm)	145-150	150-155	155-160	160-165	165-170	170-175	175-180	180-185
No. of students	2	5	9	15	16	7	5	1

Correct answer: 165

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 165.

Question 19 - Q11

In a group of 25 students 7 students failed and 6 students got distinction and the marks of remaining 12 students are 61, 36, 44, 59, 52, 56, 41, 37, 39, 38, 41, 64. Find the median marks of the whole group.

Correct answer: 41

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 41.

Question 20 - Q12

The median weight of a group of 79 students is found to be 55 kg. 6 more students are added to this group whose weights are 50, 51, 52, 59.5, 60, 61 kg. What will be the value of median of the combined group if the lowest and the highest weights were 53 kg and 59 kg respectively?

Correct answer: 55

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 55.

Question 21 - Q13

The median of the following incomplete table is 92. Find the missing frequencies x, y

C.I.	30-50	50-70	70-90	90-110	110-130	130-150	Total
f	6	x	18	20	y	10	80

Options

- A. 14, 12
- B. 12, 14
- C. 16, 10
- D. 10, 16

Correct answer: 14, 12

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 14, 12.

Question 22 - Q14

A company produces tables which are packed in batches of 100.

An analysis of the defective tubes in different batches has received the following information given in table.

Estimate the number of defective tubes in the central batch.

No. of defective tubes	Less than 5	5-9	10-14	15-19	20-24	25-29	30 and above
No. of batches	45	51	84	39	20	8	4

Correct answer: 11

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 11.

Question 23 - Q15

In a college there are 500 students in junior college, 5% score less than 25 marks, 68 score from 26 to 30 marks, 30% score from 31 to 35 marks, 70 score from 36 to 40 marks, 20% score from 41 to 45 marks and the rest score 46 and above marks.

What is the median marks?

Correct answer: 36

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 36.

Question 24 - Q16

Draw a cumulative frequency curve more than type for the following data and hence locate Q_1 and Q_3 .

Daily wages more than (₹)	100	150	200	250	300	350	400	450	500
No. of workers	200	188	160	124	74	49	31	15	5

Options

A. $(Q_1, Q_3) = (220, 340)$

B. $(Q_1, Q_3) = (210, 330)$

C. $(Q_1, Q_3) = (230, 350)$

D. $(Q_1, Q_3) = (200, 320)$

Correct answer: $(Q_1, Q_3) = (220, 340)$

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is $(Q_1, Q_3) = (220, 340)$.

Question 25 - Q16(i)

Draw a cumulative frequency curve more than type for the following data, after locating Q_1 and Q_3 , find the number of workers with daily wages between ₹170 and ₹260.

Daily wages more than (₹)	100	150	200	250	300	350	400	450	500
No. of workers	200	188	160	124	74	49	31	15	5

Correct answer: 62

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 62.

Question 26 - Q16(ii)

Draw a cumulative frequency curve more than type for the following data, after locating Q_1 and Q_3 , find the number of workers with daily wages less than ₹260.

Daily wages more than (₹)	100	150	200	250	300	350	400	450	500
No. of workers	200	188	160	124	74	49	31	15	5

Correct answer: 86

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 86.

Question 27 - Q17

Draw ogive of both the types for the following frequency distribution and hence find median.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No. of students	5	5	8	12	16	15	10	8	5	2

Correct answer: 48

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 48.

Question 28 - Q18(i)

Find Q_1 for the following data :

C.I.	8-8.95	9-9.95	10-10.95	11-11.95	12-12.95
f	5	10	20	10	5

Correct answer: 9.725

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully. The verified result is 9.725.

Question 29 - Q18(ii)

Find D_6 for the following data :

C.I.	8-8.95	9-9.95	10-10.95	11-11.95	12-12.95
f	5	10	20	10	5

Correct answer: 10.725

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 10.725.

Question 30 - Q18(iii)

Find P_{78} for the following data :

C.I.	8-8.95	9-9.95	10-10.95	11-11.95	12-12.95
f	5	10	20	10	5

Correct answer: 11.375

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 11.375.

Question 31 - Q19

For below data, find all quartiles

Weight (kg)	40-45	45-50	50-55	55-60	60-65	65-70	70-75	75-80
No. of persons	4	15	20	30	20	10	8	4

Options

- A. $(Q_1, Q_2, Q_3) = (52.1875, 57.75, 63.5625)$
- B. $(Q_1, Q_2, Q_3) = (50.1875, 55.75, 61.5625)$
- C. $(Q_1, Q_2, Q_3) = (53.1875, 58.75, 64.5625)$
- D. $(Q_1, Q_2, Q_3) = (51.1875, 56.75, 62.5625)$

Correct answer: $(Q_1, Q_2, Q_3) = (52.1875, 57.75, 63.5625)$

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is $(Q_1, Q_2, Q_3) = (52.1875, 57.75, 63.5625)$.

Question 32 - Q19

For below data, find number of persons weighing between 57 kg and 72 kg.

Weight (kg)	40-45	45-50	50-55	55-60	60-65	65-70	70-75	75-80
No. of persons	4	15	20	30	20	10	8	4

Correct answer: 51

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 51.

Question 33 - Q20

For the following data showing weights of 100 employees,
find the maximum weight of the lightest 25% of employees.

Weight (kg)	45-50	50-55	55-60	60-65	65-70	70-75	75-80
No. of employees	6	8	15	26	20	14	11

Correct answer: 58.67

Explanation and steps

Choose the quartile, decile, percentile or ogive method required by the question and substitute the table values carefully.
The verified result is 58.67.